

TTDS 2023 Past Paper Deep Solutions

Original-question guide + worked solutions

Generated for TTDS revision

2026-05-20

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使用方式

这份解析对应 2023638_INFR11145_INFR11229.pdf。每题都用“题目要点 + 解题思路 + 答案/计算 + 常见坑 + 考试模板”组织。中文解释为主，关键 TTDS terms 保留英文，方便考试直接写。

Question 1: Short Answers

Original question screenshot

Please provide answers to **ALL** of the following questions. It is more important to have a clear answer than a longer answer. When calculations are needed, please **show steps** and give answer to **three** decimal digits.

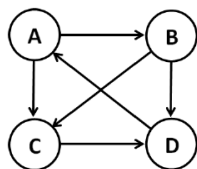
1. Provide a short answer to the following questions: [10 marks]

- What are the components of a “test collection” in the Cranfield paradigm?
- Why is smoothing important in language modelling?
- What does the PageRank value, as discussed in the lecture, of a given web page represent?
- Why don't commercial web search engines typically provide explicit relevance feedback functionality?
- What is meant by a “shingle”?
- What is meant by a “polite” crawler?
- Can “eye tracking” be used as implicit feedback? If so, how?
- What are the three major categories of user needs in Web Search?
- How can an automatic search system detect that two queries are similar if they don't share a single word?
- What is the role of the back queues in the crawler URL frontier?

2. Consider the following web pages and their out-links: [5 marks]

- Page A points to page B and C.
- Page B points to page C and D.
- Page C points to D.
- Page D points to page A.

Compute the PageRank scores at the end of the first two iterations for each of the pages. Teleportation factor is 0.5. Show your steps.



3. In the past few months, chatGPT and similar AI chat-bots have shown impressive performance. Some opinions claim that these chat bots will replace the web search engines. Do you agree with this opinion? Discuss in clear points why you agree or disagree. Justify your points with examples if possible. [5 marks]

原题要点

10 个 short-answer concepts: Cranfield test collection、smoothing、PageRank、explicit relevance feedback、shingle、polite crawler、eye tracking as implicit feedback、web search user needs、similar queries without shared words、crawler back queues。

逐题解析

(a) Cranfield paradigm 的 test collection components

标准答案是三件套: document collection、information needs / queries / topics、relevance judgements / qrels。它们让不同 IR systems 可以在同一数据、同一 queries、同一 relevance labels 下比较。

(b) Why smoothing matters in language modelling

在 query likelihood language model 中, 如果 query term 没在 document 出现, MLE 会给 $P(t|d) = 0$, 导致整个 query probability 变成 0。Smoothing 把 document model 和 collection model 混合, 既避免 zero probability, 也让短文档的估计更稳。

(c) PageRank value represents what

PageRank 是 random surfer 在长期随机浏览中位于某 page 的 stationary probability。直觉上, 它是 link-based importance, 不直接看 query content。

(d) Why commercial web search rarely has explicit relevance feedback

用户不愿额外标注 relevant/non-relevant, 交互成本高, 而且 web search 常是快速任务。商业搜索更常用 implicit feedback, 如 clicks、dwell time、query reformulation。

(e) Shingle

Shingle 是 document 中连续 token/character 的 k -gram, 用于 near-duplicate detection。比如 word 3-shingles 可表示局部文本片段, 再用 Jaccard/MinHash 比较相似度。

(f) Polite crawler

Polite crawler 不会过度请求同一 host, 会遵守 robots.txt、crawl-delay、per-host waiting time, 避免给网站服务器造成压力。

(g) Eye tracking as implicit feedback

可以。若用户长时间注视某 result/snippet/document, 可能表示 attention 或 interest。但它 noisy: 看了不等于 relevant, 也可能是困惑。

(h) Three major web search user needs

经典分类: informational、navigational、transactional。Informational 是找信息, navigational 是去特定网站, transactional 是完成购买、下载、订票等动作。

(i) Detect similar queries without shared words

可用 query logs、click-through data、embeddings/dense representations、thesaurus/query expansion。如果两个 queries 点击同一组 documents, 或 dense vectors 接近, 即使没有 shared terms 也可判为 similar。

(j) Role of back queues in URL frontier

Back queues 通常按 host/site 管理 URLs, 保证同一 host 的 fetch 间隔满足 politeness。Front queues 偏 prioritisation, back queues 偏 scheduling/politeness。

考试模板

Short-answer 不要写散文。格式可用: Term = definition + why it matters + one caveat/example。

Question 2: PageRank Computation

Original question screenshot

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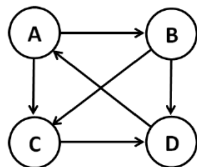
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Compute the PageRank scores at the end of the first two iterations for each of the pages. Teleportation factor is 0.5. Show your steps.



3. In the past few months, chatGPT and similar AI chat-bots have shown impressive performance. Some opinions claim that these chat bots will replace the web search engines. Do you agree with this opinion? Discuss in clear points why you agree or disagree. Justify your points with examples if possible. [5 marks]

原题要点

Graph:

- A → B, C
- B → C, D
- C → D
- D → A

Teleportation factor / damping factor 按讲义公式理解为 $\lambda = 0.5$, 即:

$$PR_{i+1}(x) = \frac{1-\lambda}{N} + \lambda \sum_{y \rightarrow x} \frac{PR_i(y)}{L_{out}(y)}$$

这里 $N = 4$, 初始 $PR_0(A) = PR_0(B) = PR_0(C) = PR_0(D) = 0.25$, teleportation base 是 $(1-0.5)/4 = 0.125$ 。

计算步骤

Iteration 1:

- $PR_1(A) = 0.125 + 0.5PR_0(D) = 0.125 + 0.125 = 0.250$
- $PR_1(B) = 0.125 + 0.5PR_0(A)/2 = 0.125 + 0.0625 = 0.188$
- $PR_1(C) = 0.125 + 0.5(PR_0(A)/2 + PR_0(B)/2) = 0.250$
- $PR_1(D) = 0.125 + 0.5(PR_0(B)/2 + PR_0(C)) = 0.313$

Iteration 2:

- $PR_2(A) = 0.125 + 0.5PR_1(D) = 0.281$
- $PR_2(B) = 0.125 + 0.5PR_1(A)/2 = 0.188$
- $PR_2(C) = 0.125 + 0.5(PR_1(A)/2 + PR_1(B)/2) = 0.234$
- $PR_2(D) = 0.125 + 0.5(PR_1(B)/2 + PR_1(C)) = 0.297$

答案

Page	Iteration 1	Iteration 2
A	0.250	0.281
B	0.188	0.188
C	0.250	0.234
D	0.313	0.297

常见坑

不要把 page 的 PageRank 全部给每条 outlink; 要除以 out-degree。也不要忘记 teleportation base。

Question 3: Will ChatGPT Replace Web Search

Original question screenshot

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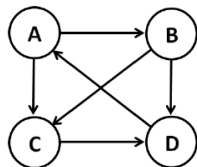
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原题要点

要求讨论 LLM/chatbot 是否会取代 web search engines。不是纯观点题，应该用 TTDS concepts 支撑。

深入解析

一个平衡答案通常比绝对同意/反对更稳：LLMs 会改变 search interface，但不会完全替代 search engines。

为什么不会完全替代：

- Freshness: web search 可以抓取最新网页, LLM parametric knowledge 可能过时。
- Attribution: search results 可给 URLs/snippets, 用户能追溯 source; LLM 可能 hallucinate。
- Navigational/transactional needs: 用户常想去官网、买票、下载文件, search engine 更直接。
- Corpus coverage: search engine 的 index 可覆盖 massive web collection, LLM 不能把所有实时网页都放进 parameters。
- Evaluation and trust: 对医疗、法律、新闻等 high-stakes queries, 需要 sources 和 ranking diversity。

LLMs 会带来的改变:

- 对 informational queries, LLM 可 summarise、compare、explain。
- RAG 可以结合 retrieval 和 generation: 先 retrieve evidence, 再 generate grounded answer。
- Chat interface 更适合 exploratory search 和 multi-turn clarification。

考试模板

我会写: LLMs complement rather than fully replace web search. Search engines provide freshness、coverage、source attribution and navigational access; LLMs provide synthesis and conversational interaction. Best system is often RAG-like: retrieval for evidence, generation for explanation.

Question 4: IR Evaluation Metrics

Original question screenshot

4. **IR Evaluation** Search engine X produced the following ranked lists for queries A,B,C,D,E,F,G,H. Here a + means that a relevant document was found at the corresponding rank, while - means that a non-relevant document was found. Assume that all relevant documents were retrieved in the shown ranked lists for the 8 queries.

rank:	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A:	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
B:	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	+	-	-
C:	+	-	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
D:	+	-	+	-	-	+	-	-	-	+	-	-	-	-	-	-	-	-	-	-
E:	-	-	-	-	-	-	+	+	+	+	-	-	+	+	+	+	-	-	-	-
F:	-	-	-	-	-	-	-	-	-	-	+	+	+	+	+	+	+	+	+	+
G:	-	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
H:	-	-	+	+	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

- (a) Compute the F1 score at rank 8 for query D. Report two decimal digits. [1 mark]
- (b) Compute the R-precision for each of the 8 queries. [3 marks]
- (c) Let nDCG' be a modified version of normalised Discounted Cumulative Gain (nDCG), which applies a linear discount to the gain instead of $\log_2$ in the original formula, so that no discount is applied at rank 1, and discount keeps increasing linearly to contribute only 10% of the document's relevance at rank 10, then no contribution at all for documents ranked below 10..

i. Write down the formula for your definition of DCG' [2 marks]

ii. Assuming "+" stands for gain = 1, and "-" stands for gain = 0. Calculate your nDCG' value at rank 10 for query D. [3 marks]

iii. Explain what kind of search applications might find this formula more useful than the original ones. [1 mark]

原题要点

8 个 queries A-H，每个 ranked list 用 +/- 表示 relevant/non-relevant。要算：

- Q4(a) query D 的 F1@8
- Q4(b) 每个 query 的 R-precision
- Q4(c) 自定义线性折扣的 nDCG'@10

(a) F1@8 for Query D

Query D relevant ranks: 1, 3, 6, 10, 所以总 relevant 数 $R = 4$ 。

Top 8 中 relevant 有 3 个:

$$P@8 = \frac{3}{8} = 0.375, \quad R@8 = \frac{3}{4} = 0.750$$

$$F1@8 = \frac{2PR}{P+R} = \frac{2(0.375)(0.750)}{0.375 + 0.750} = 0.500$$

(b) R-precision

R-precision = P@R, 其中 R 是该 query 的 relevant documents 总数。

Query	Relevant ranks	R	R-precision
A	1	1	1.000
B	1-18, 20	19	0.947
C	1, 3	2	0.500
D	1, 3, 6, 10	4	0.500
E	7, 8, 9, 10, 13, 14, 15, 16	8	0.250
F	11-20	10	0.000
G	2, 3	2	0.500
H	3, 4	2	0.000

(c) Modified nDCG'

题目定义 linear discount: rank 1 weight = 1, rank 10 weight = 0.1, rank > 10 weight = 0。可定义:

$$DCG'@10 = \sum_{i=1}^{10} rel_i (1 - 0.1(i-1))$$

Query D relevant ranks: 1, 3, 6, 10。

$$DCG'@10 = 1 + 0.8 + 0.5 + 0.1 = 2.4$$

Ideal ranking 把 4 个 relevant 放在 ranks 1-4:

$$IDCG'@10 = 1 + 0.9 + 0.8 + 0.7 = 3.4$$

$$nDCG'@10 = \frac{2.4}{3.4} = 0.706$$

什么时候 modified formula 更有用?

适合用户几乎只看 top 10, 且 rank 10 之后价值近似为 0 的场景, 如 web search first page、mobile search、answer recommendation。它比 log discount 更 harshly penalise late relevant documents。

Question 5: Expected Mutual Information

Original question screenshot

5. You have collected a set of newspaper articles. Each article (document) belongs to exactly one category, such as *Germany* or *drink*. You have identified a number of words that are of interest to you, such as *London* and *match*, and counted how many documents in each category they appear in:

	UK	Germany	food	drink	politics	entertainment
Total no. of docs	1024	1216	564	702	1020	605
London	204	43	45	34	120	40
Berlin	12	302	51	43	79	37
Britain	561	69	56	45	204	120
broccoli	43	45	122	90	5	3
Indonesia	32	51	64	56	105	140
voters	202	198	2	12	504	7
match	67	67	12	12	89	321

Calculate the expected mutual information $I(U; C)$ for the word *broccoli* and the category *food* (show steps).

$$\begin{aligned}
 I(U; C) &= \sum_{e_t \in \{1,0\}} \sum_{e_c \in \{1,0\}} P(U = e_t, C = e_c) \log_2 \frac{P(U = e_t, C = e_c)}{P(U = e_t)P(C = e_c)} \\
 &= \frac{N_{11}}{N} \log_2 \frac{NN_{11}}{N_1.N_1} + \frac{N_{01}}{N} \log_2 \frac{NN_{01}}{N_0.N_1} + \frac{N_{10}}{N} \log_2 \frac{NN_{10}}{N_1.N_0} + \frac{N_{00}}{N} \log_2 \frac{NN_{00}}{N_0.N_0}
 \end{aligned}$$

[5 marks]

6. Provide a short answer to the following questions: [5 marks]

- What is the value of $I(U; C)$ if term and class are completely independent?
 - What is the main limitation of probabilistic Latent Semantic Indexing (pLSI) compared with Latent Dirichlet Allocation (LDA)?
 - In Gibbs sampling for LDA, how do we assign a new topic to a word?
 - What is feature scaling?
 - In text classification, why do we need three separate datasets for training, development and testing?
7. In 2017, a news report appeared with the title “Google’s Sentiment Analyzer Thinks Being Gay Is Bad.” It describes a machine learning-based sentiment analysis system released by Google. The problem is that this tool was found to be biased. Sentences would receive different sentiment ratings based on which demographic characteristics or identity terms (such as “gay”) were mentioned in them, even when humans agreed that the sentiment in the sentences was the

QUESTION CONTINUES ON NEXT PAGE

原题要点

计算 word *broccoli* 与 class *food* 的 expected mutual information。

从表中：

- Total $N = 1024 + 1216 + 564 + 702 + 1020 + 605 = 5131$
- $N_{11} = 122$: *broccoli* 且 *food*
- $N_{1.} = 43 + 45 + 122 + 90 + 5 + 3 = 308$: *broccoli* 总出现 doc 数

- $N_{.1} = 564$: food doc 数
- $N_{10} = 308 - 122 = 186$
- $N_{01} = 564 - 122 = 442$
- $N_{00} = 5131 - 122 - 186 - 442 = 4381$

计算

公式：

$$I(U; C) = \frac{N_{11}}{N} \log_2 \frac{N N_{11}}{N_{1.} N_{.1}} + \frac{N_{01}}{N} \log_2 \frac{N N_{01}}{N_{0.} N_{.1}} + \frac{N_{10}}{N} \log_2 \frac{N N_{10}}{N_{1.} N_{.0}} + \frac{N_{00}}{N} \log_2 \frac{N N_{00}}{N_{0.} N_{.0}}$$

代入：

Cell	Count	Contribution
N_{11}	122	0.043974
N_{01}	442	-0.022599
N_{10}	186	-0.020287
N_{00}	4381	0.025037

$$I(U; C) = 0.026125 \approx 0.026$$

常见坑

MI 不是只看 N_{11} 。一个 word 即使在某 class 里出现不少，如果它在其他 classes 也常出现，MI 不一定高。

Question 6: More Short Answers

Original question screenshot

5. You have collected a set of newspaper articles. Each article (document) belongs to exactly one category, such as *Germany* or *drink*. You have identified a number of words that are of interest to you, such as *London* and *match*, and counted how many documents in each category they appear in:

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QUESTION CONTINUES ON NEXT PAGE

Page 3 of 4

(a) MI under independence

如果 term 和 class 完全 independent, 则 $P(U, C) = P(U)P(C)$, log ratio 为 0, 所以:

$$I(U; C) = 0$$

(b) pLSI limitation vs LDA

pLSI 没有完整 generative model for new documents, 参数数量随 documents 增长, 容易 overfit。LDA 加入 Dirichlet priors, 把 document-topic distribution 作为 random variable, 因此更适合 generalise to unseen documents。

(c) Gibbs sampling in LDA

对某个 word token, 先 remove 当前 topic assignment, 再根据 conditional posterior 重新采样 topic:

$$P(z_i = k | \dots) \propto (n_{d,k}^{-i} + \alpha) \frac{n_{k,w}^{-i} + \beta}{n_k^{-i} + V\beta}$$

直觉: topic 要同时适合该 document, 也要适合该 word。

(d) Feature scaling

Feature scaling 把 features 放到相近 range, 如 standardisation 或 min-max scaling。它避免大尺度 features 在 SVM/logistic regression/kNN 等模型里支配学习。

(e) Why train/dev/test

- training set: fit model parameters
- development/validation set: tune hyperparameters、select features、choose model
- test set: final unbiased evaluation

如果用 test set 调参, test performance 会被污染, 失去真实泛化估计。

Question 7: Bias in Sentiment Classifier

Original question screenshot

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QUESTION CONTINUES ON NEXT PAGE

QUESTION CONTINUED FROM PREVIOUS PAGE

same. For example, the system rated the sentence “I’m Christian” as slightly positive but the sentence “I’m a Sikh” as very positive, even though humans would rate both sentences as neutral.

[5 marks]

- (a) Explain, using your knowledge from TTDS, why you think this issue may have occurred.
 - (b) Come up with two alternative approaches that could be used to remedy this problem. For each of the two, discuss their advantages and disadvantages.
8. Your colleague is building a text classifier for detecting misinformation on social media. Her system will identify posts that will later be reviewed by human content moderators. She knows that her classifier won’t work perfectly, so she accepts if some posts that do not contain misinformation are identified. But it is very important for her that all posts that might contain misinformation are flagged by the algorithm, so they can be identified. She shows you some of her data:
- Misinformation: “Rio de Janeiro is the capital of Japan”
 - Not misinformation: “Tokyo is the capital of Japan”
 - Misinformation: “Joe Biden was born in Japan” “They are a really strong team who are doing really well”
 - Not misinformation: “Obama was born in Hawaii.”
- (a) How would you suggest she evaluate how well her classifier is doing? Recommend an evaluation metric and justify your choice.
 - (b) Give an example of a test sentence that her approach is likely to work on, and one that it is unlikely to work on. Justify your choice.

[5 marks]

原题要点

Google sentiment system 对 identity terms 给出不同 sentiment，即使 human labels 应该 neutral。问原因和 remedy。

为什么会发生？

可能原因：

- Training data bias: identity terms 在训练语料中常与 negative/positive contexts 共现，模型学到了 spurious correlation。
- Representation bias: word embeddings 或 features 把 demographic terms 与 sentiment-bearing words 联系起来。

- Label bias: annotators 或 historical data 本身带有社会偏见。
- Domain mismatch: model 在普通 sentiment data 上训练，但被用于 identity statements，语义不同。
- Bag-of-words limitation: 如果模型主要看 words 而不理解 sentence pragmatics, I am X 可能被某个 identity token 主导。

Remedy 1: Counterfactual data augmentation

构造 pairs，如 I am Christian / I am Sikh / I am gay，保持 label 不变，只替换 identity term。

优点：直接针对 fairness bug，能减少 spurious identity-sentiment association。

缺点：覆盖不完所有 identities 和 contexts，也可能过度平滑真实语义差异。

Remedy 2: Bias-aware evaluation and regularisation

建立 fairness benchmark，测 counterfactual invariance；训练时加入 penalty，要求 identity swap 后 sentiment score 不应大幅变化。

优点：不仅修一个词，而是形成 evaluation + training discipline。

缺点：需要额外 labels/benchmarks，且 fairness metrics 可能和 raw accuracy trade off。

可补充方案

人工审核高风险 outputs、domain-specific retraining、remove/protect sensitive attributes。但简单删除 identity terms 会损失语义，也不能解决 proxy bias。

Question 8: Misinformation Classifier Evaluation

Original question screenshot

QUESTION CONTINUED FROM PREVIOUS PAGE

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原题要点

目标是 flag 可能包含 misinformation 的 posts，后续给 human moderators。可以接受 false positives，但非常不希望漏掉 misinformation。

(a) 推荐 evaluation metric

应优先用 Recall for misinformation class:

$$Recall = \frac{TP}{TP + FN}$$

因为任务的核心是找出所有可能 misinformation, false negative 代价高。也可以报告 F_{β} with $\beta > 1$, 如 F_2 , 让 recall 权重大于 precision。实际系统还应看 precision, 避免 moderators 被过多 false positives 淹没。

(b) Likely works / fails examples

Likely works:

- “Paris is the capital of Japan.”

理由: 结构接近训练中的 false factual claim, 实体关系简单, capital-country mismatch 明显。

Likely fails:

- “They are doing a really excellent job and everyone knows the numbers are fake.”

理由: 需要 context、speaker intent、world knowledge 或 discourse understanding。Bag-of-words classifier 可能被 positive words 或 vague wording 干扰。

考试模板

For high-recall moderation systems, optimise recall or F_{β} with $\beta > 1$ because missing harmful/misinformation content is worse than sending extra items to human review. Still monitor precision for moderator workload.

Coverage Checklist

Question	Covered TTDS topics
Q1	Cranfield paradigm, smoothing, PageRank, relevance feedback, shingles, crawler politeness, implicit feedback, web user needs, query similarity, URL frontier
Q2	PageRank, random surfer, teleportation/damping, iterative graph computation
Q3	Web search, LLMs, RAG-style retrieval plus generation, freshness, attribution
Q4	Precision, recall, F1, R-precision, DCG/nDCG, rank discounting
Q5	Expected mutual information, contingency table, feature selection
Q6	MI independence, pLSI vs LDA, Gibbs sampling, feature scaling, train/dev/test
Q7	Text classification bias, sentiment analysis, counterfactual evaluation, mitigation
Q8	Classification evaluation, recall, precision, F_{β} , moderation use case